THE UNIVERSITY OF AUCKLAND

SEMESTER ONE 2024
Campus: City

COMPUTER SCIENCE

AI Architecture and Design

MID-SEMESTER TEST
16th of April 2024

Time Allowed: 60 minutes (5 minutes reading time + 55 min answering time)

NOTES:

- This test is out of **30** marks.
- There are 7 short-answer questions. Attempt **ALL** questions.
- This is a restricted book test. You are only allowed to use a **double-sided page of notes**.
- Digital devices, including calculators and phones are not permitted.
- Write your answers directly after each question.
- Cross out notes and other work you do not wish to be assessed.
- We privilege quality over quantity, i.e., you do not need to write very long answers. Be concise and clear.
- Write as clearly as possible, in plain English.

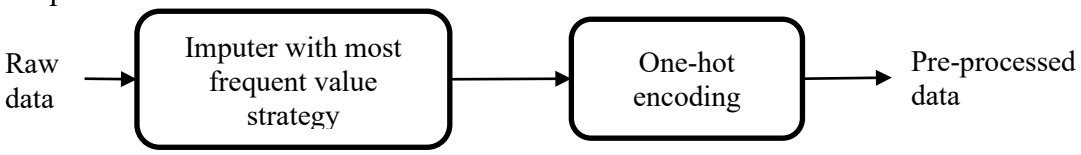
Answer ALL questions.

Question 1: Data preprocessing **[4 marks]**

Consider the 2 following preprocessing pipelines. For each pipeline, describe likely characteristics of the input raw data and explain how you can derive them from the preprocessing steps.

Pipeline 1:

Pipeline 2:



Question 2: Design choices **[6 marks]**

You trained a simple neural network with the following settings: 5 hidden layers with ReLU units, 20 neurons per layer, random initialisation, trained for 1000 epochs. The training accuracy is 95% and the validation accuracy is 60%. For each of the following suggestions, explain if it is likely to improve the validation accuracy and why, if implemented in the next iteration of the training.

- Training for 2000 epochs with the same dataset.
- Training the model on a larger dataset, including additional collected data.
- Using L2 regularisation in each layer.

Question 3: Evaluation **[4 marks]**

Consider the following confusion matrix, resulting from the evaluation of a model on a test set containing 1000 instances.

		True class	
		Positive	Negative
Predicted class	Positive	500	400
	Negative	0	100

Calculate the accuracy, precision, and recall. What do you think about this model's performance?

Question 4: Learning rate and optimisers **[4 marks]**

1) Give an example of a learning rate schedule and explain briefly why it might be beneficial when training a Deep Neural Network (DNN).
2) Explain in your own words what is the momentum mechanism and what is its effect on the gradient descent optimisation process.

Question 5: RNN and Transformer **[4 marks]**

The RNN architecture processes each input in a sequence sequentially.

1) Explain why this is at the same time an advantage and a drawback of this architecture.
2) Explain how the Transformer architecture processes inputs (on the encoder side) to alleviate this drawback of RNNs.

Question 6: CNN feature map **[4 marks]**

Answer the following questions about feature map dimensions on CNN.

1) What will be the dimensions of the feature map at the output of a convolutional layer using ten $5 \times 5 \times 5$ filters, a stride of 3 and a padding of 0, knowing that the input dimension is $50 \times 50 \times 5$? Show your calculation steps.
2) What will be the dimensions of the feature map at the output of an average pooling layer using a 5×5 filter and a stride of 5, knowing that the input dimension is $50 \times 50 \times 5$? Show your calculation steps.
3) Same question as 2), this time using a max pooling layer.

Question 7: DNN training **[4 marks]**

1) Explain why using neural networks with a lot of layers and weights per layer make the optimisation process harder.
2) Give two possible strategies which make the training of deep neural networks easier and briefly explain their effect on the training.